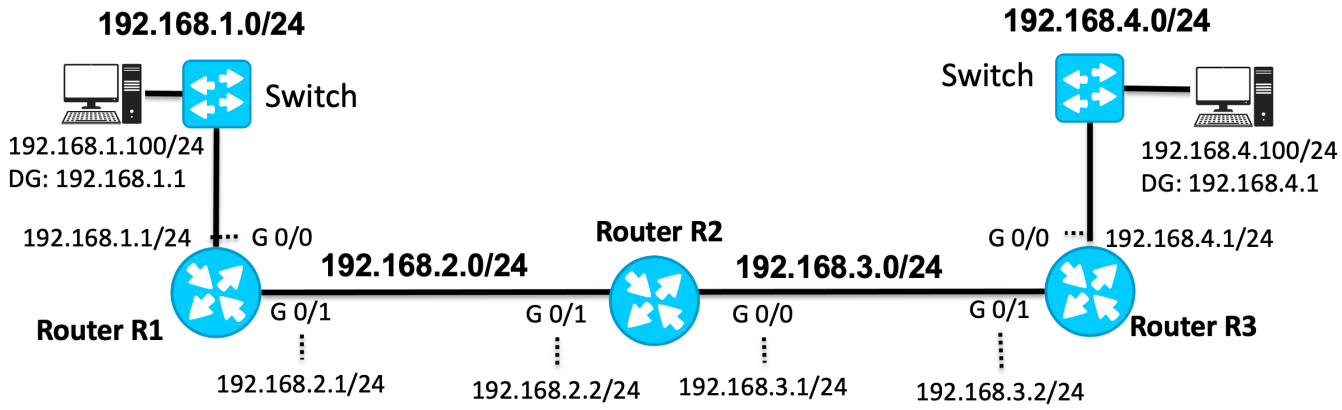
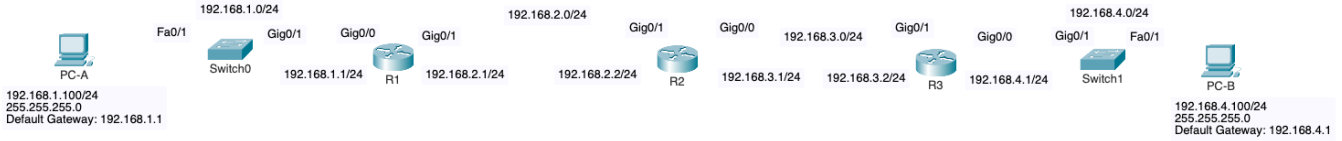
Static Routing Lab - Student Notes

Network Topology



Packet Tracer Startup Topology



Configure router R1

Method 1: Type in the commands using CLI

```
! Change from user mode to privileged exec mode
Router> enable

! Enter global configuration mode
Router# conf t

! Disable DNS lookup in case you mistype a command
Router(config)# no ip domain-lookup

! Configure the hostname
Router(config)# hostname R1

! Configure console port
R1(config)# line console 0
R1(config-line)# logging synchronous  ! Keeps output from interrupting input
R1(config-line)# exec-time 0 0   ! Keep device from timing out (lab only)
R1(config-line)# password cisco  ! Password required when connecting to console port
R1(config-line)# login           ! Login prompt
R1(config-line)# exit            ! Exit current mode (line mode)
R1(config)#
```

```
R1(config)# banner motd $        ! Banner message of the day, using $ as delimiter
Enter TEXT message.  End with the character '$'.
############################
Authorized Access Only!
############################
$

! Password required for entering privileged exec mode
R1(config)# enable secret class
R1(config)#
R1(config)# exit     ! Exit current mode (global config)

! Save running-config (RAM) to startup-config (NVRAM)
R1# copy running-config startup-config

Verification commands

R1# show running-config
R1# show startup-config
```

What if I make a mistake in IOS?

Don't worry if you make a mistake while configuring a Cisco device; IOS provides a simple way to correct it. Most commands can be undone by using the "**no**" keyword followed by the original command at the appropriate prompt.

For example, if you accidentally set an enable secret password and need to remove it:

```
R3(config)# enable secret class
R3(config)# no enable secret class
```

Similarly, if you assign an incorrect IP address to an interface, you can remove it like this:

```
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# no ip address 192.168.1.1 255.255.255.0
```

Some commands are specifically used to reverse default settings. For instance:

- To prevent the router from attempting DNS lookups when you mistype a command:
  `R1(config)# no ip domain-lookup`

- To bring an interface out of the default "shutdown" state and enable it
  `R1(config-if)# no shutdown`

By understanding how to reverse commands, you'll be able to troubleshoot and correct configurations efficiently.

Part 3: Configure Interfaces on R1, R2, and R3

Step 1. Examine the Interfaces

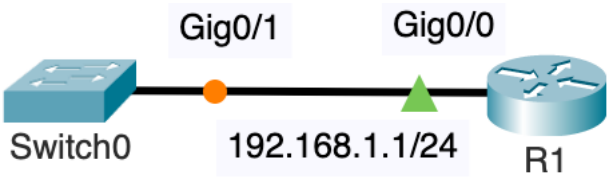
```
R1# show ip interface brief
Interface                  IP-Address      OK? Method Status                Protocol
Embedded-Service-Engine0/0 unassigned      YES unset  administratively down down
GigabitEthernet0/0         unassigned      YES unset  administratively down down
GigabitEthernet0/1         unassigned      YES unset  administratively down down
R1#
```

Step 2. Configure R1 G0/0 LAN Interface

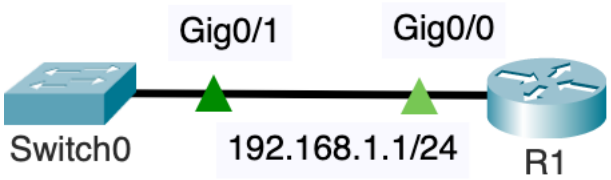
```
R1# conf t
R1(config)# interface gig 0/0
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# end
```

```
R1# show ip interface brief
Interface                  IP-Address      OK? Method Status                Protocol
Embedded-Service-Engine0/0 unassigned      YES unset  administratively down down
GigabitEthernet0/0         192.168.1.1     YES manual down                  down
GigabitEthernet0/1         unassigned      YES unset  administratively down down
R1#
```

Step 3. Connect R1's LAN G 0/0 Interface to Switchport



Link lights during STP convergence.



Link lights after STP after STP convergence.

In this step, you will physically connect R1's LAN interface (G0/0) to an Ethernet switch, enabling network connectivity.

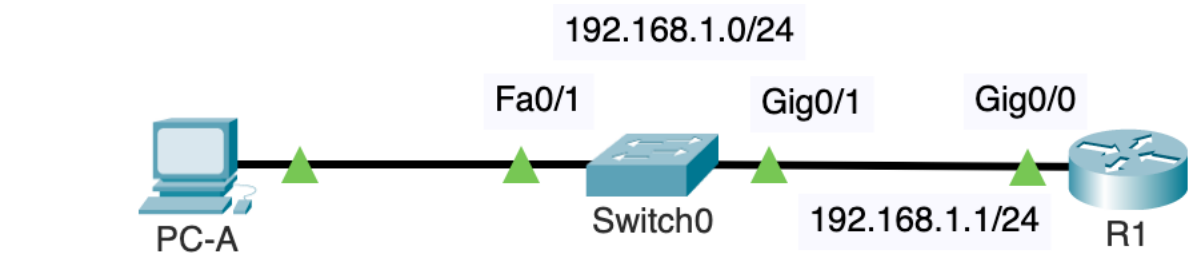
1. **Cable R1 to the LAN Switch**:
2. **Auto-Negotiation of Speed and Duplex**:
3. **Verify the Connection**:

```
*Apr 21 13:38:55.387: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to up
*Apr 21 13:38:56.387: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0,
changed state to up
R1#
```

Step 4. Verify the interface is now up

```
R1# show ip interface brief
Interface                  IP-Address      OK? Method Status                Protocol
Embedded-Service-Engine0/0 unassigned      YES unset  administratively down down
GigabitEthernet0/0         192.168.1.1     YES manual up                    up
GigabitEthernet0/1         unassigned      YES unset  administratively down down
R1#
```

Step 5: Connect and Configure Client Computer PC-A

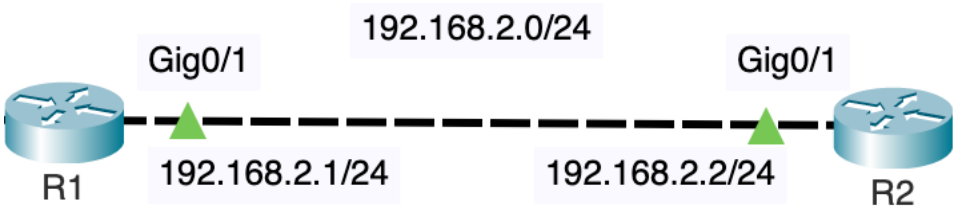


192.168.1.100/24
255.255.255.0
Default Gateway: 192.168.1.1

Link lights after STP after STP convergence.

1. **Cable PC-A to Switch**:
2. **Configure IP Settings on PC-A**:
    - **IPv4 Address**:
    - **Subnet Mask**:
    - **Default Gateway**:

Step 6. Configure and Enable R1's G0/1 Inter-router Interface with R2



Link lights after both R1 and R2 have been configured in step 6 and step 7.

1. **Cable R1 to R2**:
2. **Configure R1's G0/1 Interface**:

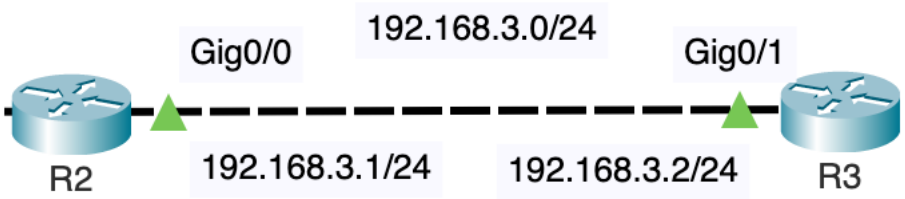
```
R1(config)# inter g 0/1
R1(config-if)# ip add 192.168.2.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# exit
```

Step 7: Configure R2's G 0/1 inter-router interface with R1

1. **Configure R2's G0/1 Interface**:

```
R2(config)# inter g 0/1
R2(config-if)# ip add 192.168.2.2 255.255.255.0
R2(config-if)# no shutdown
R2(config-if)# exit
```

Step 8: Configure R2's G 0/0 inter-router interface with R3 G 0/1



Link lights after both R1 and R2 have been configured in step 8 and step 9.

1. **Cable R2 to R3**:
2. **Configure R2's G0/0 Interface**:

```
R2(config)# inter g 0/0
R2(config-if)# ip add 192.168.3.1 255.255.255.0
R2(config-if)# no shutdown
R2(config-if)# exit
```

Step 9: Configure R3's G 0/1 inter-router interface with R2

1. **Configure R3's G0/1 Interface**:

```
R3(config)# inter g 0/1
R3(config-if)# ip add 192.168.3.2 255.255.255.0
R3(config-if)# no shutdown
R3(config-if)# exit
```

Step 10. Configure R3's G0/0 LAN Interface

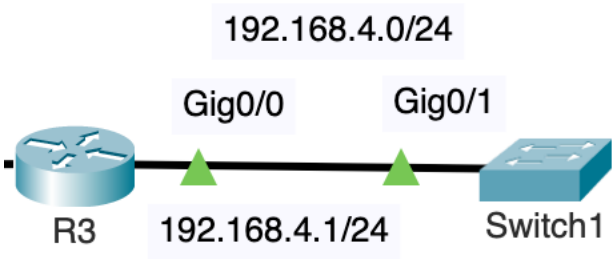
1. **Configure R3's G0/0 Interface**:

```
R3# conf t
R3(config)# interface gig 0/0
R3(config-if)# ip address 192.168.4.1 255.255.255.0
R3(config-if)# no shutdown
R3(config-if)# end
```

Note on Interface Status:

- After configuring the interface, you might notice that the interface status is still down. This is because the interface is not yet connected to an Ethernet switch or any other device.

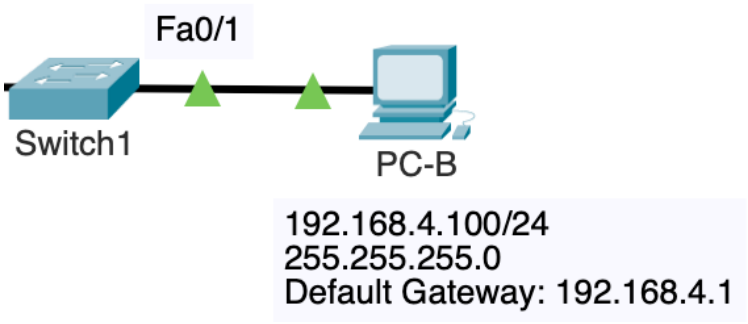
Step 11. Connect R3's LAN G 0/0 Interface to Switchport



Link lights after step 10 and step 11.

1. **Cable R3 to the LAN Switch**:
2. **Ensure the Switch is Powered On**:
3. **Purpose**:

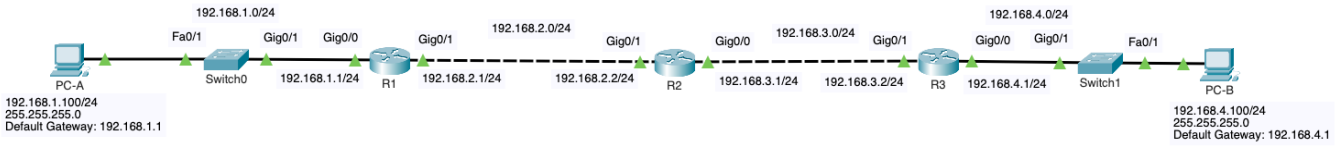
Step 12: Configure and Connect Client Computer Option



Link lights after STP after STP convergence.

1. **Cable PC-B to Switch1**:
2. **Configure IP Settings on PC-B**:
    - **IPv4 Address**:
    - **Subnet Mask**:
    - **Default Gateway**:

Completed topology



Your topology is now completed and should look like the one above.

Step 13. Verify on all three routers that their GigabitEthernet interfaces are now up

```
R1# show ip inter brief
Interface                  IP-Address      OK? Method Status                Protocol
Embedded-Service-Engine0/0 unassigned      YES unset  administratively down down
GigabitEthernet0/0         192.168.1.1     YES manual up                    up
GigabitEthernet0/1         192.168.2.1     YES manual up                    up
R2#

R2# show ip inter brief
Interface                  IP-Address      OK? Method Status                Protocol
Embedded-Service-Engine0/0 unassigned      YES unset  administratively down down
GigabitEthernet0/0         192.168.3.1     YES manual up                    up
GigabitEthernet0/1         192.168.2.2     YES manual up                    up
R2#

R3# show ip inter brief
Interface                  IP-Address      OK? Method Status                Protocol
Embedded-Service-Engine0/0 unassigned      YES unset  administratively down down
GigabitEthernet0/0         192.168.4.1     YES manual up                    up
GigabitEthernet0/1         192.168.3.2     YES manual up                    up
R2#
```

Part 4: Testing Reachability

Step 1: Examine the routing tables of R1, R2 and R3.

1. **Check R1's Routing Table**:

```
R1# show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
<output omitted>

Gateway of last resort is not set

      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.1.0/24 is directly connected, GigabitEthernet0/0
L        192.168.1.1/32 is directly connected, GigabitEthernet0/0
      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.2.0/24 is directly connected, GigabitEthernet0/1
L        192.168.2.1/32 is directly connected, GigabitEthernet0/1
R1#
```

2. **Check R2's Routing Table**:

```
R2# show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
<output omitted>

Gateway of last resort is not set

      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.2.0/24 is directly connected, GigabitEthernet0/1
L        192.168.2.2/32 is directly connected, GigabitEthernet0/1
      192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.3.0/24 is directly connected, GigabitEthernet0/0
L        192.168.3.1/32 is directly connected, GigabitEthernet0/0
R2#
```

3. **Check R3's Routing Table**:

```
R3# show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
<output omitted>

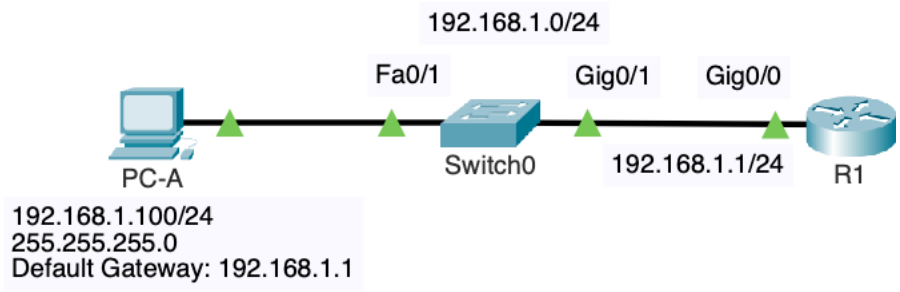
Gateway of last resort is not set

      192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.3.0/24 is directly connected, GigabitEthernet0/1
L        192.168.3.2/32 is directly connected, GigabitEthernet0/1
      192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.4.0/24 is directly connected, GigabitEthernet0/0
L        192.168.4.1/32 is directly connected, GigabitEthernet0/0
R2#
```

Step 2. Test Reachability from End Devices

From PC-A

Can PC-A ping its default gateway, router R1?




```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

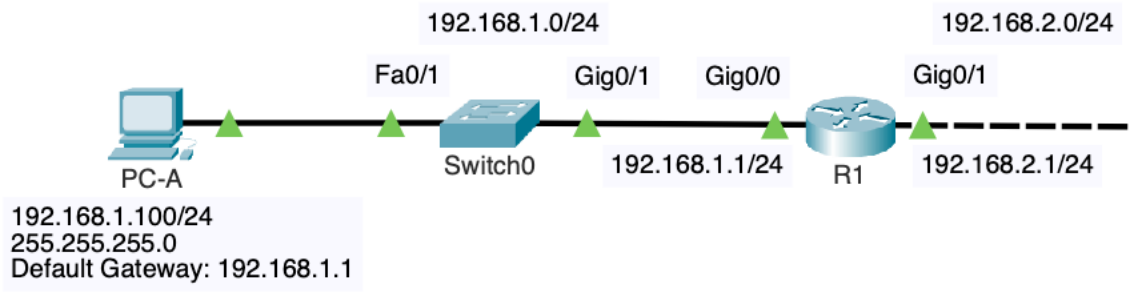
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
```

```
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Can PC-A ping the other interface on router R1?

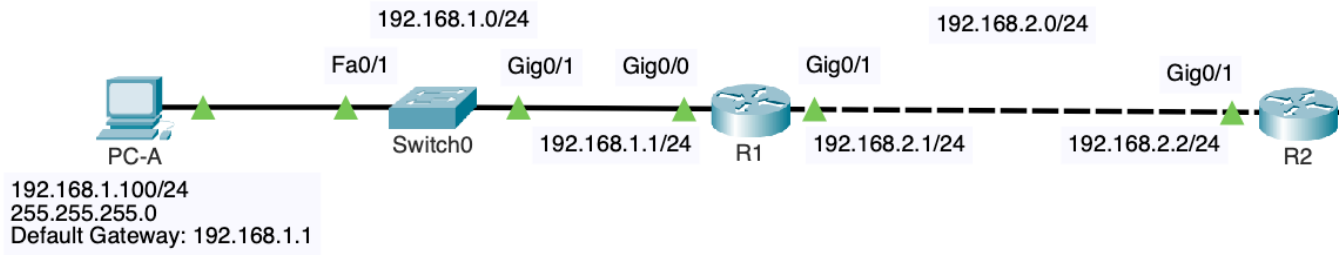



```
C:\>ping 192.168.2.1

Pinging 192.168.2.1 with 32 bytes of data:

Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
<output omitted>
```

Can PC-A ping the closest interface on router R2?




```
C:\> ping 192.168.2.2

Pinging 192.168.4.100 with 32 bytes of data:

Request timed out.
Request timed out.
<output omitted>
```

Step 3: Verify that R1 and R2 can ping each other on same interconnected network

1. **Ping from R1 to R2**:

```
R1# ping 192.168.2.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.2.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/1 ms
R1#
```

2. **View the ARP Table on R1**:
   - After the ping, you can view the ARP table on R1 to see the resolved MAC addresses for the connected devices.

```
R1# show arp
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  192.168.1.1              -   64f6.9d1c.8970  ARPA   GigabitEthernet0/0
Internet  192.168.2.1              -   64f6.9d1c.8971  ARPA   GigabitEthernet0/1
Internet  192.168.2.2              1   64f6.9d1c.8c41  ARPA   GigabitEthernet0/1
R1#
```

3. **Ping from R2 to R1**:

```
R2# ping 192.168.2.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.2.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (4/5), round-trip min/avg/max = 1/1/1 ms
R1#
```

Step 4: Verify that R2 and R3 can ping each other on same interconnected network

1. **Ping from R2 to R3**:

```
R2# ping 192.168.3.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/1 ms
R1#
```

2. **Ping from R3 to R2**:

```
R3# ping 192.168.3.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (4/5), round-trip min/avg/max = 1/1/1 ms
R1#
```

Step 5: Verify routers cannot reach Remote networks

From R1

```
R1# ping 192.168.3.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.1, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
R1#

R1# ping 192.168.4.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.4.1, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
R1#
```

From R2: Unable to reach remote networks

```
R2# ping 192.168.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.1, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
R2#

R2# ping 192.168.4.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.4.1, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
R2#
```

From R3: Unable to reach remote networks

```
R3# ping 192.168.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.1, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
R2#

R3# ping 192.168.2.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.2.1, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
R2#
```

Part 5: Configuring Static Routes

Step 1. Configure Static Routes on R1 and R2 to reach remote networks

Remember Alex Zinin's Routing Table Principles:

- **Principle 1:** Every router makes its decision alone, based on the information it has in its own routing table.
- **Principle 2:** The fact that one router has certain information in its routing table does not mean that other routers have the same information.
- **Principle 3:** Routing information about a path from one network to another does not provide routing information about the reverse, or return, path.

Configuring Static Routes on R1

1. **To Reach the 192.168.3.0/24 Network**:

```
R1# conf t
R1(config)# ip route 192.168.3.0 255.255.255.0 192.168.2.2
R1(config)#
```

2. **To Reach the 192.168.4.0/24 Network**:

```
R1# conf t
R1(config)# ip route 192.168.4.0 255.255.255.0 192.168.2.2
R1(config)#
```

Configuring Static Routes on R2

1. **To Reach the 192.168.1.0/24 Network**:

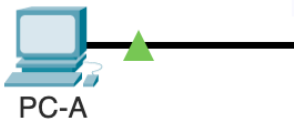
```
R2# conf t
R2(config)# ip route 192.168.1.0 255.255.255.0 192.168.2.1
R2(config)#
```

2. **To Reach the 192.168.4.0/24 Network**:

```
R2# conf t
R2(config)# ip route 192.168.4.0 255.255.255.0 192.168.3.2
R2(config)#
```

Troubleshooting Connectivity from PC-A to PC-B

Attempt to ping PC-B from PC-A.



192.168.1.100/24
255.255.255.0
Default Gateway: 192.168.1.1

```
C:\> ping 192.168.4.100

Pinging 192.168.4.100 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Reply from 192.168.1.1: Destination host unreachable.

Ping statistics for 192.168.4.100:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

The ping from PC-A to PC-B fails. Why? Consider the following:

1. **Is the packet reaching its destination?**

2. **Is there an issue with the return path of the packets?**

Step 2. Configure Static Routes on R3 to reach remote networks

R3's Static Routes Configuration

1. **To Reach the 192.168.2.0/24 Network**:

```
R3# conf t
R3(config)# ip route 192.168.2.0 255.255.255.0 192.168.3.1
R3(config)#
```

2. **To Reach the 192.168.1.0/24 Network**:

```
R3# conf t
R3(config)# ip route 192.168.1.0 255.255.255.0 192.168.3.1
R3(config)#
```

Step 3: Verify Static Routes in IP Routing Tables

1. **Check R1's Routing Table**:

```
R1# show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
<output omitted>

Gateway of last resort is not set

      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.1.0/24 is directly connected, GigabitEthernet0/0
L        192.168.1.1/32 is directly connected, GigabitEthernet0/0
      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.2.0/24 is directly connected, GigabitEthernet0/1
L        192.168.2.1/32 is directly connected, GigabitEthernet0/1
S     192.168.3.0/24 [1/0] via 192.168.2.2
S     192.168.4.0/24 [1/0] via 192.168.2.2
R1#
```

2. **Check R2's Routing Table**:

```
R2# show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
<output omitted>

Gateway of last resort is not set

S     192.168.1.0/24 [1/0] via 192.168.2.1
      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.2.0/24 is directly connected, GigabitEthernet0/1
L        192.168.2.2/32 is directly connected, GigabitEthernet0/1
      192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.3.0/24 is directly connected, GigabitEthernet0/0
L        192.168.3.1/32 is directly connected, GigabitEthernet0/0
S     192.168.4.0/24 [1/0] via 192.168.3.2

R2#
```

3. **Check R3's Routing Table**:

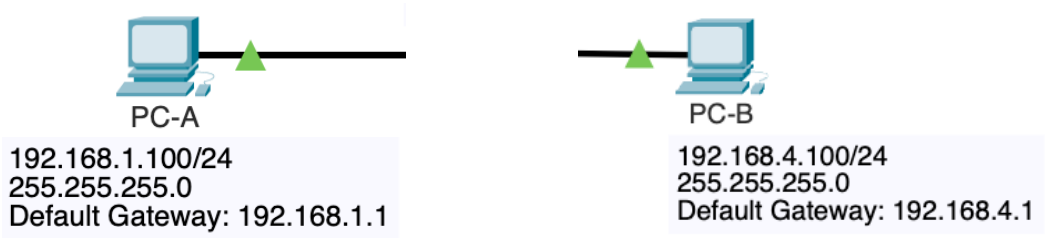
```
R3# show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
<output omitted>

Gateway of last resort is not set

S     192.168.1.0/24 [1/0] via 192.168.3.1
S     192.168.2.0/24 [1/0] via 192.168.3.1
      192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.3.0/24 is directly connected, GigabitEthernet0/1
L        192.168.3.2/32 is directly connected, GigabitEthernet0/1
      192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.4.0/24 is directly connected, GigabitEthernet0/0
L        192.168.4.1/32 is directly connected, GigabitEthernet0/0
R2#
```

Step 4. Verify reachability from PC-A

1. **Ping from PC-A to PC-B**:




```
C:\>ping 192.168.4.100

Pinging 192.168.4.100 with 32 bytes of data:

Reply from 192.168.4.100: bytes=32 time<1ms TTL=125
Reply from 192.168.4.100: bytes=32 time<1ms TTL=125
Reply from 192.168.4.100: bytes=32 time<1ms TTL=125
Reply from 192.168.4.100: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.4.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

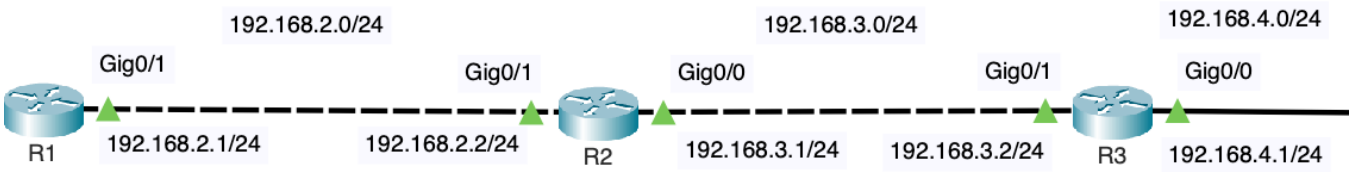
C:\>
```

Step 5. Verify reachability to remote networks from router R1

1. **Ping Remote Networks from R1**:




```
R1# ping 192.168.3.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/4 ms
R1#
```




```
R1# ping 192.168.4.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.4.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/4 ms
R1#
```

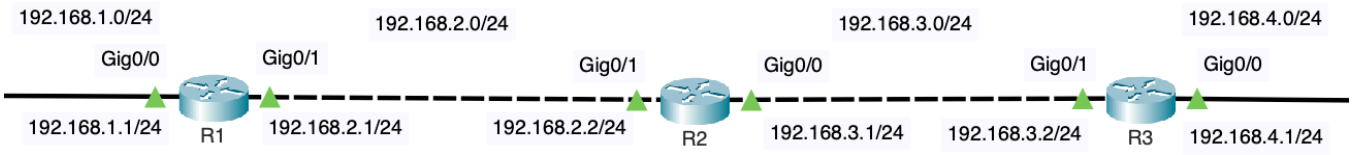
Step 6. Sending a ping from a different router interface

Sending a Ping with a Specified Source Address

1. **Specify the Source Interface**:

Example:

- To send a ping from R1's GigabitEthernet0/0 interface (IP address 192.168.1.1) to the IP address 192.168.4.1:



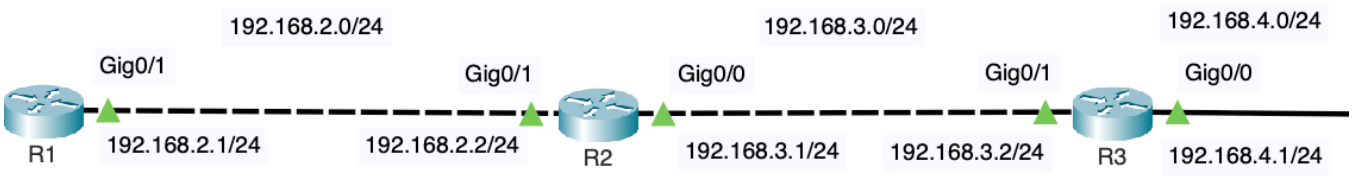

```
R1# ping 192.168.4.1 ?
  data      specify data pattern
  df-bit    enable do not fragment bit in IP header
  repeat    specify repeat count
  size      specify datagram size
  source    specify source address or name
  timeout   specify timeout interval
  tos       specify type of service value
  validate  validate reply data
  <cr>

R1# ping 192.168.4.1 source gig 0/0
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.4.1, timeout is 2 seconds:
Packet sent with a source address of 192.168.1.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R1#
```

Step 7. Use Traceroute to see the path the packets take

Traceroute Examples

1. **Traceroute from R1 to 192.168.4.1**:

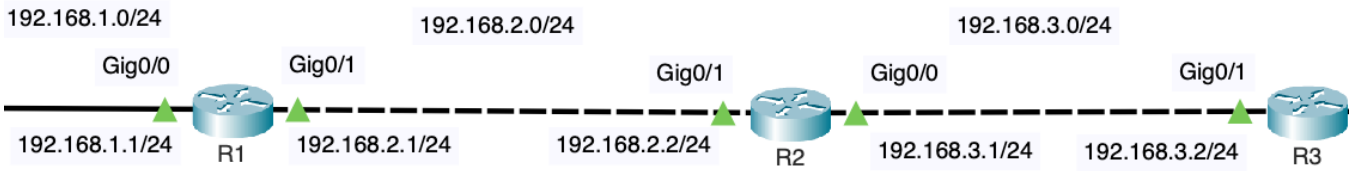



```
R1# traceroute 192.168.4.1

Type escape sequence to abort.
Tracing the route to 192.168.4.1

  1 192.168.2.2  0 msec  0 msec  0 msec
  2 192.168.3.2  4 msec 4 msec 4 msec
```

2. **Traceroute from R3 to 192.168.1.1**:




```
R3# traceroute 192.168.1.1

Type escape sequence to abort.
Tracing the route to 192.168.1.1

  1 192.168.3.1  0 msec  0msec  0 msec
  2 192.168.2.1  4 msec 4 msec 4 msec
```

Congratulations! You've successfully completed the lab!